

Chapter 30 Induction and Inductance

Induction Experiment

When the **magnetic flux through the loop changes**, that is, when the magnetic field and the loop have relative motion or when the magnetic field through the loop changes, a **current will be generated in the loop**—**induced current**, indicating that an electromotive force—**induced emf**, is generated in the loop.

Faraday's Law

The total induced emf. of N-turn coils

$$\epsilon = -N \frac{d\Phi_B}{dt}$$

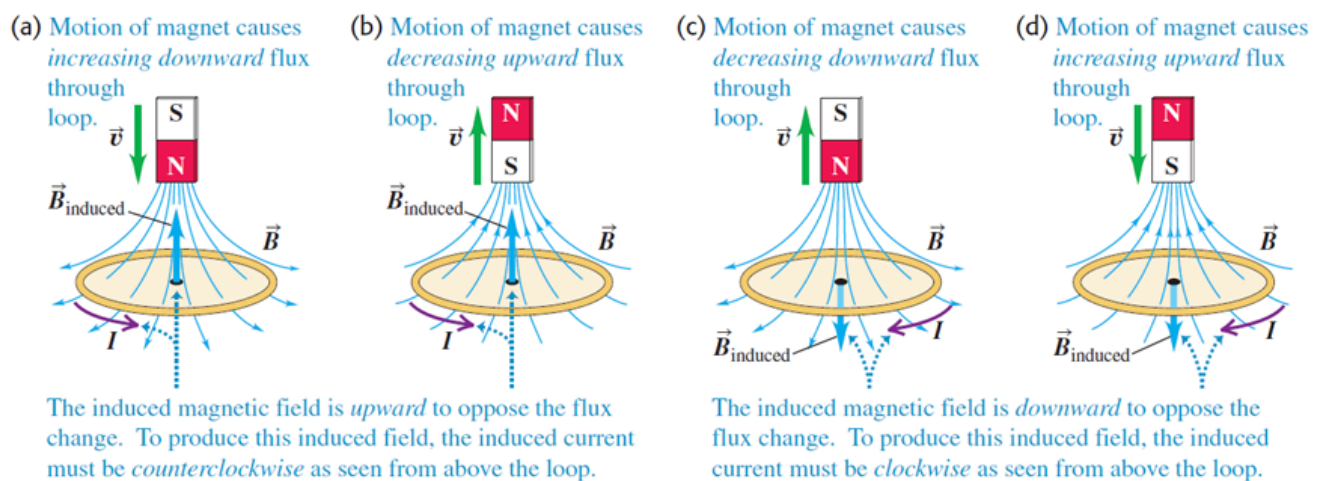
General methods to change the magnetic flux:

- Change the magnitude of B
- Change the valid area S

Lenz's Law

The direction of any magnetic induction effect is such as to **oppose the cause of the effect**.

Actually a manifestation of the principle of **conservation of energy**.



Use **right hand**:

- The fingers are in the current's direction

- The thumb is the induced fields direction